# Fine-Tuning vs. RAG vs. Prompt Engineering + Hybrid Architectures

## Three Strategies to Tailor an LLM

When you want an LLM to perform a specific task better, you have three main tools:

- **Prompt Engineering:** Craft better instructions in the prompt — no training required.
- **RAG (Retrieval-Augmented Generation):** Inject live, retrieved documents into the prompt at inference time.
- **Fine-Tuning:** Re-train the model on task-specific data to bake new behaviour into the weights.

## Comparison: Mechanism, Best Use Case, Cost/Complexity

| | Prompt Engineering | RAG | Fine-Tuning |
|---|---|---|---|
| **Mechanism** | Better instructions in context | Retrieve & inject docs | Update model weights |
| **Best For** | General tasks, formatting | Live/private knowledge | Style, domain specialisation |
| **Data Needed** | None | Document corpus | Labelled task examples |
| **Latency** | Lowest | Medium (retrieval step) | Lowest (post-training) |
| **Cost** | Near zero | Storage + retrieval | GPU compute + time |
| **Complexity** | Low | Medium | High |

## Hybrid Architectures in Production

Most production AI systems don't use just one strategy — they combine all three:

- Fine-tune the base model on company-specific tone and domain vocabulary.
- Add RAG to inject real-time product documentation and knowledge base articles.
- Use prompt engineering to format output, control persona, and handle edge cases.

> *Example: A customer service bot is fine-tuned on past support tickets (style), uses RAG for live product docs (knowledge), and uses prompt engineering to enforce response format.*

## Decision Framework — Which to Use?

- Need current/private data? → RAG first.
- Need specific writing style or domain expertise baked in? → Fine-tuning.
- Need quick improvement with no infrastructure? → Prompt engineering.
- Need all three? → Hybrid architecture.

## Quick-Revision Questions

- When would you choose RAG over fine-tuning?
- What is the approximate cost tradeoff between the three strategies?
- Describe a hybrid architecture for a legal research assistant.